SIMATS ENGINEERING
ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA0510** **Slot A**

COURSE OUTCOME COVERED

CO4: ***Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*** **Assessment Tool Weightage: 40%**

QUESTIONS:5 **Total Marks 50** **Duration :60 Minutes Annexure A**

Industry Problem Solving Task

Code of Conduct:

I AKHILA POCHIMIREDDY(192425225) ***certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.***

Signature of the student: Akhila.P

Rubrics for Evaluation

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	***Marks(100)***
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, welldocumented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

1. Smart Surveillance System Selection

A city surveillance system must operate under the following conditions:

- If latency > 100 ms → system fails
- If accuracy < 85% → unacceptable
- Hardware supports only moderate computation Models available:

Viola-Jones: 78%, low latency

YOLO: 92%, moderate latency

Deep Learning: 95%, high latency

Compare the models and evaluate which satisfies all constraints. Justify your decision using logical reasoning based on given conditions.

Given Conditions

- Latency **> 100 ms → system fails**
- Accuracy **< 85% → unacceptable**
- Hardware supports only **moderate computation**

Available models:

- Viola-Jones: **78% accuracy, low latency**
- YOLO: **92% accuracy, moderate latency**
- Deep Learning: **95% accuracy, high latency**

Comparison

Model	Accuracy	Latency	Computation	Meets Requirements?
Viola-Jones	78%	Low	Low	❌ No – accuracy < 85%
YOLO	92%	Moderate	Moderate	☑ **Yes**
Deep Learning	95%	High	High	❌ No – may exceed latency/hardware limits

Analysis

Viola-Jones satisfies the low-latency requirement, but its accuracy is only 78%, which is below the required 85%. Therefore, it cannot be selected.

Deep Learning provides the highest accuracy at 95%, but it has high latency and requires more computational resources. Since the hardware supports only moderate computation and latency must not exceed 100 ms, it is not the safest choice.

YOLO provides **92% accuracy**, moderate latency, and moderate computational requirements. Therefore, it satisfies the accuracy, latency, and hardware constraints.

Decision

YOLO should be selected.

It provides the best balance between accuracy, response time, and computational requirements for a city surveillance system.

2. Autonomous Vehicle

Detection Logic A vehicle detection system must:

- Prioritize safety (accuracy > 90%)
- Maintain response time < 50 ms Options:

Method A: 88% accuracy, 30 ms

Method B: 93% accuracy, 70 ms

Method C: 91% accuracy, 45 ms

Compare the methods and evaluate which should be selected. Justify your choice using constraint-based reasoning.

Given Conditions

The autonomous vehicle system must:

- Achieve **accuracy > 90%**
- Maintain response time **< 50 ms**

Methods:

- Method A: **88%, 30 ms**
- Method B: **93%, 70 ms**
- Method C: **91%, 45 ms**

Comparison

Method	Accuracy	Response Time	Accuracy Requirement	Time Requirement	Decision
A	88%	30 ms	✗	☑	Reject

B	93%	70 ms	☑	✗	Reject
C	**91%**	**45 ms**	☑	☑	**Select**

Analysis

Method A has a fast response time of 30 ms, but its accuracy is only 88%. Since autonomous vehicles require accuracy greater than 90%, Method A fails the safety requirement.

Method B has 93% accuracy, satisfying the safety requirement. However, its response time is 70 ms, which exceeds the maximum allowed 50 ms. Therefore, Method B cannot be selected.

Method C achieves 91% accuracy and a response time of 45 ms. It satisfies **both constraints simultaneously**.

Decision

Method C should be selected because it is the only method satisfying both the accuracy and response-time requirements.

This is especially important in autonomous vehicles because a highly accurate method is not sufficient if it responds too slowly to a changing road situation.

3. Motion Analysis Decision

Problem A traffic system requires:

- Accurate motion direction detection
- Robustness in dense traffic Results:

Optical Flow: accurate but noisy in dense scenes

Motion Estimation: stable but less detailed

Compare both approaches and evaluate which is suitable under high-density conditions. Justify using scenario-based logic.

Given Conditions

The traffic system requires:

- Accurate motion-direction detection
- Robustness in dense traffic

Results:

- **Optical Flow:** Accurate but noisy in dense scenes
- **Motion Estimation:** Stable but less detailed

Comparison

Factor	Optical Flow	Motion Estimation
Motion direction	Accurate	Moderate
Detail	High	Lower
Dense traffic	Noisy	**More stable**
Robustness	Lower in dense scenes	**Higher**
Suitability	Less suitable	**More suitable**

Analysis

Optical Flow provides detailed and accurate motion information. However, in dense traffic, many vehicles move close to each other. This can produce noisy or ambiguous flow vectors, reducing reliability.

Motion Estimation provides less detailed information, but its stability makes it more appropriate when many objects are present simultaneously.

Since the application specifically requires **robustness in dense traffic**, stability is more important than maximum motion detail.

Decision

Motion Estimation should be selected for high-density traffic conditions.

A hybrid system could also be considered: Motion Estimation can provide stable overall movement, while Optical Flow can be applied selectively when detailed motion information is required.

4. Background Modeling Evaluation A GMM-based system produces:

- Static scenes: 90% accuracy
- Dynamic scenes: 65% accuracy Requirement:

System must maintain ≥80% accuracy across all scenarios

Compare performance and evaluate whether GMM alone meets requirements. Justify logically and suggest a decision.

Given Data

GMM performance:

- Static scenes: **90% accuracy**
- Dynamic scenes: **65% accuracy**
- Requirement: **≥ 80% accuracy in all scenarios**

Comparison

Scenario	**GMM Accuracy**	**Required Accuracy**	**Result**
Static	90%	≥80%	☑ Meets
Dynamic	65%	≥80%	✖ Fails

Analysis

GMM performs well in static scenes, achieving 90% accuracy, which is above the required 80%. However, in dynamic scenes, accuracy drops to 65%. This is **15 percentage points below** the required minimum.

Dynamic backgrounds may contain moving trees, water, shadows, rain, or other changing elements. These changes can make background and foreground separation more difficult.

Therefore, GMM alone does **not** satisfy the system requirement because the requirement must be maintained across **all scenarios**.

Proposed Solution

A better system should combine GMM with additional techniques such as:

1. **Adaptive background updating** to handle changing environments.
2. **Morphological filtering** to remove noise.
3. **Shadow removal** to reduce false foreground detection.
4. **Object tracking** to maintain consistent object detection.
5. **Deep-learning-based object detection** for difficult dynamic scenes.

Decision

GMM alone should not be used. A hybrid background-modeling approach should be adopted to improve performance in dynamic environments and achieve the required ≥80% accuracy.

5. Depth Estimation Strategy

Constraints:

- Only one camera available
- Moderate depth accuracy acceptable Options:

Stereo Vision: high accuracy, needs two cameras

Structure from Motion: moderate accuracy, single camera

Compare both approaches and evaluate which is feasible. Justify using constraint-based reasoning.

Given Conditions

- Only **one camera** is available.
- **Moderate depth accuracy** is acceptable.

Available methods:

- **Stereo Vision:** High accuracy, requires two cameras.
- **Structure from Motion (SfM):** Moderate accuracy, requires only one camera.

Comparison

Factor	**Stereo Vision**	**Structure from Motion**
Cameras required	2	**1**
Accuracy	High	Moderate
Compatibility with one camera	✕ No	☑ **Yes**
Meets required accuracy	☑	☑
Feasibility	Low	**High**

Analysis

Stereo Vision calculates depth using images from two cameras. Although it provides high depth accuracy, it cannot be directly used because the system has only one available camera.
Structure from Motion estimates depth by analyzing the movement of a single camera across multiple frames. Its accuracy is moderate, which is sufficient because the requirement only specifies that moderate depth accuracy is acceptable.
Therefore, SfM satisfies both important constraints:

- It requires only one camera.
- It provides the required moderate depth accuracy.

Decision

Structure from Motion should be selected.

It is the most feasible solution because it satisfies the available hardware constraint while providing sufficient depth estimation accuracy.